

Adobe Interview Ques

Q1. There are N doors and a person is standing in front of every door, initially all doors are closed.

1st person toggles 1 2 3 4 5 ... N

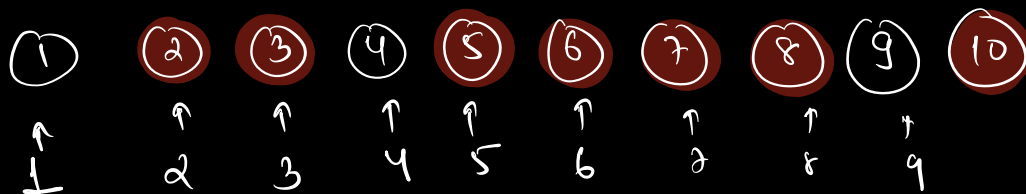
2nd person toggles 2 4 6 8 ... N

3rd person " → 3 6 9 ... N

4th person " → 4 8 12 16 ... N

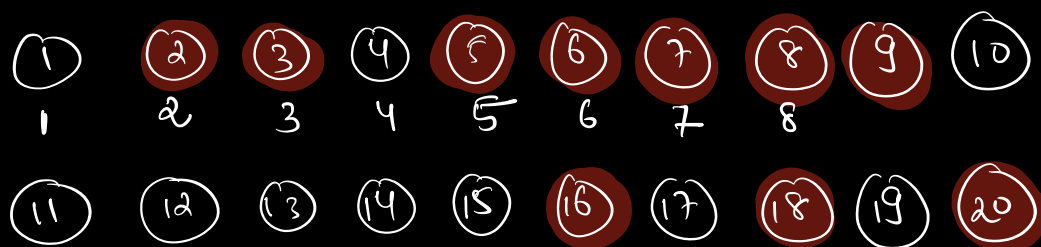
return all doors that are open in the end.

ex ⇒ $N=10$



o/p = 1 5 9

$N=20$



$$\begin{array}{ccc} 2 & \rightarrow & 1 \rightarrow 2 \\ c & & 0 \quad c \end{array}$$

$$\begin{array}{ccc} 4 & \rightarrow & 1 \rightarrow 2 \rightarrow 4 \\ c & \rightarrow & 0 \rightarrow c \rightarrow 0 \end{array}$$

$$\begin{array}{cccccc} 18 & \rightarrow & 1 & 2 & 3 & 6 & 9 & 18 \\ c & \rightarrow & 0 & \rightarrow c & \rightarrow 0 & \rightarrow c & \rightarrow 0 & \rightarrow c \end{array}$$

$$\begin{array}{ccc} 6 & \rightarrow & 1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \\ c & \rightarrow & 0 \rightarrow c \rightarrow 0 \rightarrow c \end{array}$$

$$\begin{array}{ccc} 8 & \rightarrow & 1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \\ c & \rightarrow & 0 \rightarrow c \rightarrow 0 \rightarrow c \end{array}$$

1	2	3	6	9	18
0	c	0	c	0	c
<u>✓</u>		<u>✓</u>		<u>✓</u>	
1st		3rd		5th	

odd toggle $\rightarrow$ opening -
even toggle $\rightarrow$ closing

If each factor is a toggle,
gate no. with odd ^{no. of} factors $\rightarrow$ open
gate no. with even ^{no. of} factors $\rightarrow$ closed

18

$$\begin{array}{cccccc} 1 & 2 & 3 & 6 & 9 & 18 \\ \hline \end{array}$$

$i = N/i$ ①

$$\begin{array}{ccc} i & N/i \\ \hline \text{pairs} \end{array}$$

$$25 \quad 1 \quad 5 \quad 25 \rightarrow \text{odd}$$

$$16 \quad 1 \quad 2 \quad 4 \quad 8 \quad 16 \rightarrow \text{odd}$$

$$36 \quad 1 \quad 2 \quad 3 \quad 4 \quad 6 \quad 9 \quad 12 \quad 18 \quad 36 \rightarrow \text{odd}$$

all perfect sq. have odd no. of factors
 & every other no. has even no. of factors

In the given range of doors $[1-N]$, all perfect sq.s will have open doors else all will close doors.

$$N = 100$$

$$1 \times 1 = 1$$

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

$$4 \times 4 = 16$$

⋮

⋮

⋮

$$10 \times 10 = \underline{\underline{100}}$$

$$N = 110$$

$$1 \times 1 = 1$$

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

⋮

⋮

⋮

$$11 \times 11 = \underline{\underline{121}}$$

$N \rightarrow$ doors

↓

check for each if its a perfect sq.

$$\underline{\underline{x}} \stackrel{N}{=} \sqrt{x}$$

TC $\Rightarrow N\sqrt{x} \rightarrow x$ is each no. [1 

↓

$$\underline{\underline{N\sqrt{N}}}$$

$$\underline{\underline{1-N}}$$

$\rightarrow$

$$\sqrt{x}$$

$$N \left(\sqrt{x} \right)$$

↓

$$\underline{\underline{N\sqrt{N}}}$$

Q.2. Nth magical no:-

Given a no. N , return N th magical no.

Magical no:- A no. that can be represented as a sum of
unique powers of 5, power > 0

$$1 \Rightarrow 5^1 \rightarrow 5$$

$$2 \rightarrow 5^2 \rightarrow 25$$

$$3 \rightarrow 5^1 + 5^2 \rightarrow 30$$

$$4 \rightarrow 5^3 \rightarrow 125$$

$$5 \rightarrow 5^3 + 5^1 \Rightarrow 130$$

$$6 \rightarrow 5^3 + 5^2 \Rightarrow 125 + 25 = 150$$

$$7 \rightarrow 5^3 + 5^2 + 5^1 \Rightarrow 155$$

$$8 \rightarrow 5^4 \Rightarrow 625$$

3	2	1	
0	0	5 ¹	001
0	5 ²	0	010
0	5 ²	5 ¹	011
5 ³	0	0	100
5 ³	0	5 ¹	101
5 ³	5 ²	0	110
5 ³	5 ²	5 ¹	111
			1000

$$11^{\text{th}} \text{ magical} \rightarrow \begin{matrix} 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{matrix}$$

$$5^4 + 5^2 + 5^1$$

$$= 625 + 25 + 5$$

$$= \underline{\underline{655}}$$

→ Convert 11 to binary no. 2

put power of 5, starting 5¹ rightmost position

Google

Q3. Majority element:-

Given an array of size N , return if there exists a no. with $\text{freq} > N/2$, else return -1 . { no extra space }

ex $\Rightarrow$ arr[] = 1 6 1 1 2 1 $N = 6$
 $\text{freq} \rightarrow 4 \ (4 > 6/2)$
 $\textcircled{1} \rightarrow \underline{4} > \underline{6/2}$

Q42

3 4 3 6 1 3 2 5 3 3 3
 $\Rightarrow \textcircled{3}$ ME. $N = \underline{11}$
 $11/2 = 5.5$
 $\textcircled{6}$

Q42

4 6 5 3 4 5 6 4 4 4
 $4 > N/2$ $N = 10$
O/p = $\textcircled{-1}$

Brute force

check the freq of every element in the array.
if $\text{freq} > N/2$ return n
else -1

```
for (i = 0 → N-1) {
```

```
    freq = 0
```

```
    for (j = 0 → N-1) {
```

```
        if (arr[i] == arr[j])
```

```
            freq++
```

```
    }
```

```
    if (freq > N/2)
```

```
        return arr[i]
```

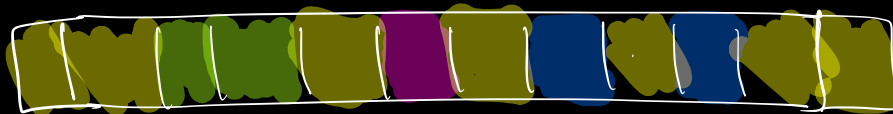
```
    }
    return -1
```

```
    }
```

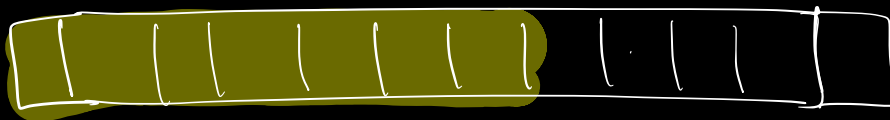
$TC \Rightarrow O(N^2)$
 $SC \Rightarrow O(1)$

∴ Can we have more than 1 majority element

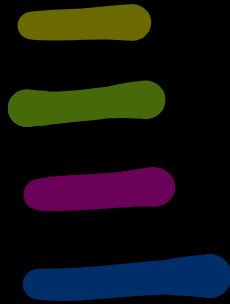
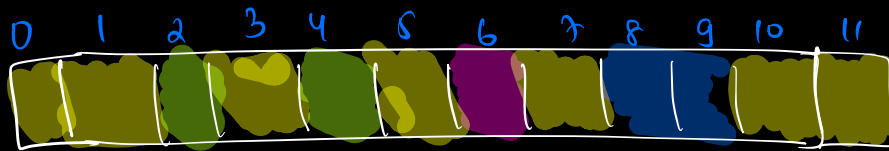
Ans
 NO NO
 XX
 NO
 false
 0.



Yellow is majority



$N = 12 \Rightarrow > N/2 \Rightarrow \underline{\underline{7}}$



yellow $\Rightarrow$ 0 1 3 5 7 10 11
 green $\Rightarrow$ 2 4
 pink $\Rightarrow$ 6
 blue $\Rightarrow$ 8 9
 → Mahagatbandhan

yellow $\Rightarrow$ 0 1 3 5 7 ~~10~~ ~~11~~ $\rightarrow$ 7 6 5

Mahagatbandhan $\Rightarrow$ 2 4 ~~6~~ ~~8~~ ~~9~~ $\rightarrow$ 5 4 3

* Count of ME will be more than combined count of all other elements.

$\Rightarrow$ If we remove two distinct elements from array

ME remains the same

ex 23

0	1	2	3	4	5	6	7	8	9	10
3	4	3	6	1	3	2	5	3	3	3

N	ME	Count ME
11	3	$6 > 11/2$
9	3	$5 > 9/2$
7	3	$5 > 7/2$
5	3	$4 > 5/2$

How to solve

Moore's Voting algorithm

3 4 3 6 1 3 2 5 3 3 3

ME $\rightarrow$ 3 3 1 2 3 3
 C $\rightarrow$ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~2~~ 3

$O(p) = 3$

3 4 3 6 1

ME $\rightarrow$ 3 3 1 $\leftarrow$
 C $\rightarrow$ ~~1~~ ~~0~~ ~~1~~ 0 1

① $\rightarrow$ for ① ~~$> 6/2$~~
 return 1

$a \Rightarrow$

0	1	2	3	4	5	6	7	8	9	10
3	3	4	3	2	2	2	3	3	3	1

 $N = 11$
 $ME = 3$

$ME \rightarrow 1$ 4 2
 $c \rightarrow$ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~2~~ ~~1~~ ~~0~~ 3]

3
 $\downarrow$ $\rightarrow$ verify
 $freq \Rightarrow \underline{6} > 11/2$ $\checkmark \rightarrow$ return 3

3	1	3	3	3	5
---	---	---	---	---	---

 $\downarrow$
 $ME \rightarrow$ 3 3 3
 $c \rightarrow$ ~~1~~ ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~2~~

$freq = 4 > 6/2$
 $\checkmark$
 $\downarrow$
return 3

1) Initially $ME = arr[0]$
 $c = 1$

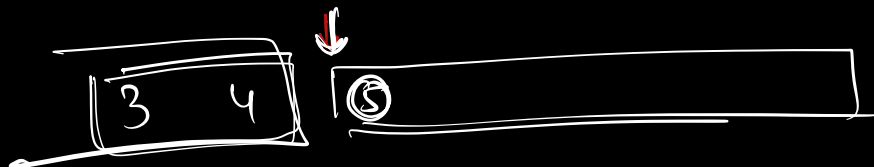
11) if $arr[i]$ matches ME
 $c++$
 else $c--$

iii) if $C=0$,

new ME = arr[i]

$C=1$

iv) take ME $\rightarrow$ last comes out & verify



ME = 3

$C = X$ 0

5

1

TC $\Rightarrow O(N+N) \Rightarrow O(N)$

$\downarrow$

find

$\downarrow$

verify

To get best case ME

SC $\Rightarrow O(1)$

HW

pseudo code

HW

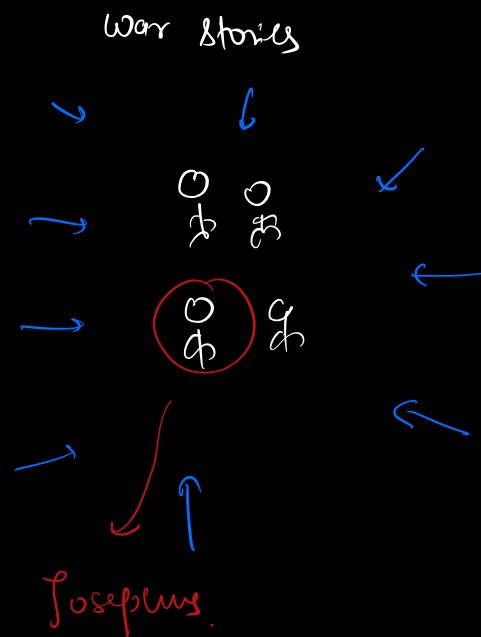
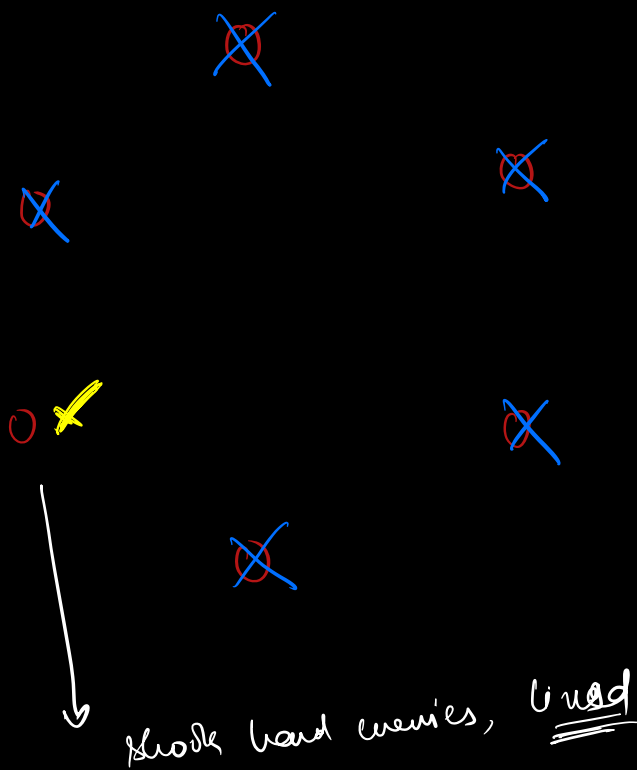
Find majority element (freq $> N/3$)

or there can be two MEs

{ no extra
space }

TC $\Rightarrow O(N)$

Q. 4. Josephus Problem



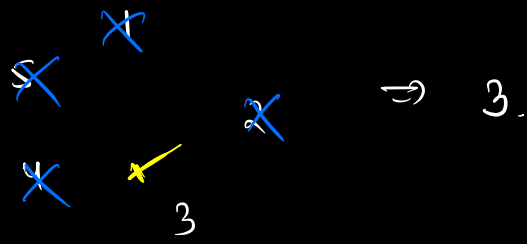
If there are N people, what should be the position to stand to be the last man standing.

$$N = 1 \longrightarrow 1$$

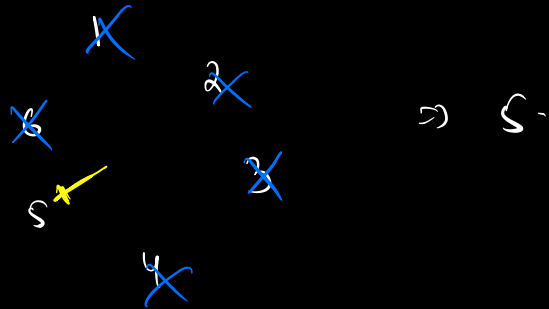
$$N = 2 \xrightarrow{\text{yellow arrow}} 1 \text{ (survives), } 2 \text{ (eliminated)} \longrightarrow 1$$

$$N = 3 \xrightarrow{\text{yellow arrow}} 3 \text{ (survives), } 1 \text{ (eliminated), } 2 \text{ (eliminated)} \longrightarrow 3$$

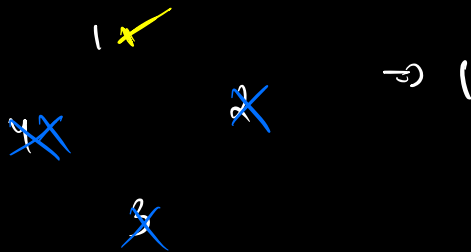
$$N = 5$$



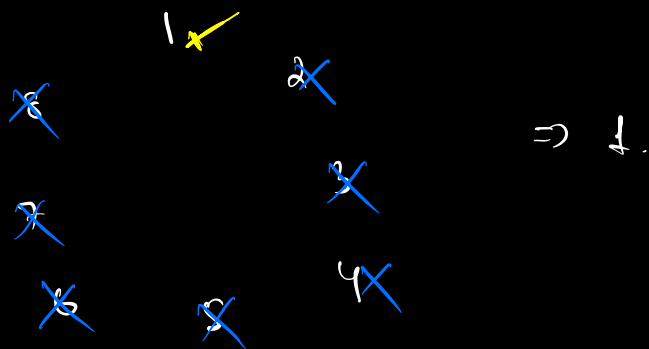
$$N = 6$$



$$N = 4$$



$$N = 8$$



$$N = 16 \Rightarrow$$

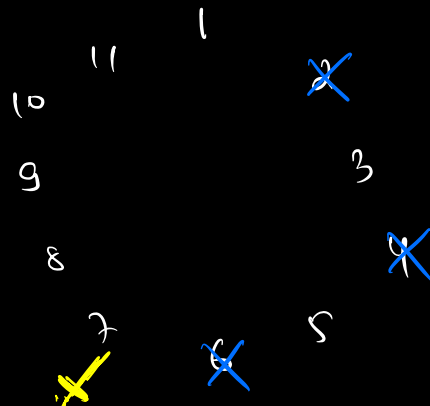
①

N		last man standing [st. from 1]
2	→	1
4	→	1
8	→	1
16	→	1
32	→	1
64	→	1

for nos. which power of 2, $2^k \rightarrow N$,

(last man standing $\rightarrow$ starting pt.)

$$\underline{\underline{N=11}}$$



when at 7,

no. of ppl alive

$$= 8.$$

↓

$$2^3$$

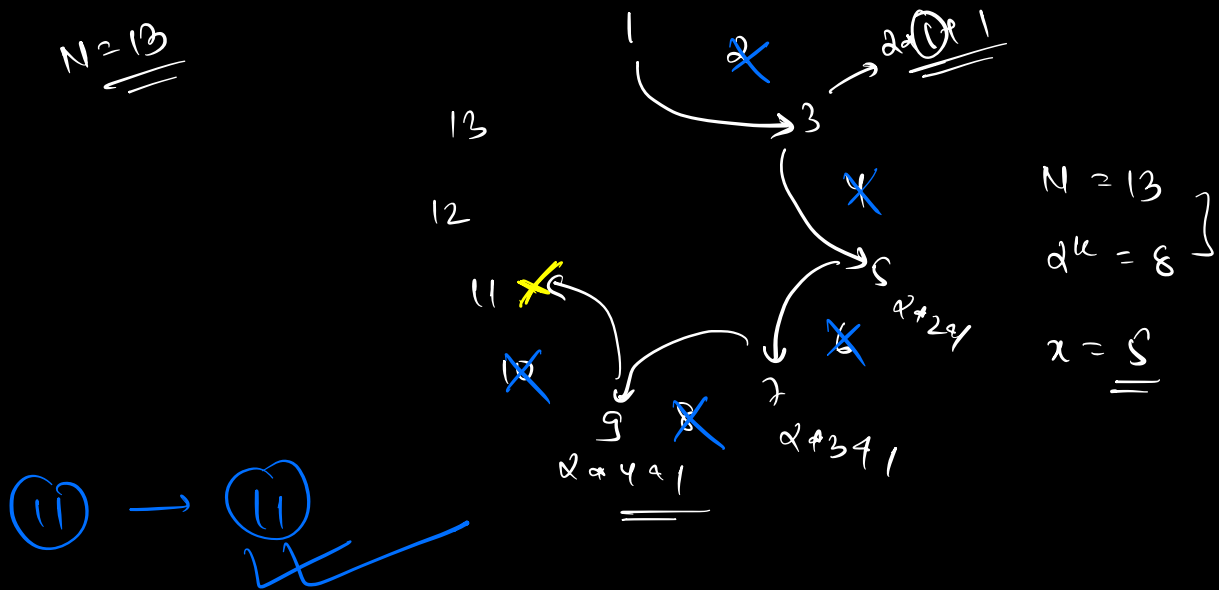
so, st. for 23

7

↓

last man standing

$$N = 13$$



Reach a no, which 2^k .

With x no. of people to reach 2^k .

$$N - 2^k = x$$

$$position = \underline{2x + 1}$$



i) find the nearest power of 2 from N ($\leq N$)

ii) $x = N - 2^k$ (nearest power of 2)

iii) $position = \underline{2x + 1}$

The Invitation Game

Game Ending

0
1 2 3 4 5 6 7 8 9
✓ ✓
10 11 12 13 14 15 16 17
18 19 20

$$\underline{\underline{i \times i \leq N}}$$

$$4 \times 4 = 16$$